

Streamflow Uncertainty Analysis under Climate and Land-Use Change

Results highlights — integrated climate / land-use / hydrological uncertainty assessment for a monsoon-dominated rain-fed river basin

Conducted as part of a larger hydrology and climate-risk consultancy assessment. Full code and methodology: see accompanying GitHub repository.

This project quantifies how much future streamflow in a monsoon-dominated, rain-fed river basin (~6,500–7,000 km², spanning steep headwaters to lowland Terai-type plains) is expected to change under climate and land-use change through 2100, and — critically — how much of that projected change is genuinely uncertain versus how much is driven by which specific climate model or land-use scenario is assumed.

35 yrs

observed daily climate & discharge baseline (1985–2020)

3

gauging stations, upstream → downstream, sequentially calibrated

8

representative CMIP6 GCM×SSP combinations, screened from 28

24

integrated future scenarios per channel (4 GCMs × 2 SSPs × 3 horizons)

2–1000 yr

flood design-discharge return periods estimated

1 Approach

- **Hydrological model:** semi-distributed SWAT+ model, calibrated upstream→downstream with Sobol global sensitivity analysis (23 parameters) and Dynamically Dimensioned Search optimization.
- **Climate ensemble:** 3-step CMIP6 screening (seasonal cycle → annual bias → four-corner envelope) narrowed 28 GCMs to 8 representative Hot-Wet / Hot-Dry / Cool-Wet / Cool-Dry combinations under SSP2-4.5 and SSP5-8.5.
- **Bias correction:** robust empirical quantile mapping for precipitation, linear (mean+variance) transformation for temperature.
- **Land-use change:** CA-Markov / neural-network transition modelling projected land cover from 2000–2020 observations forward to 2100, validated against an independent year (histogram accuracy 90%, location accuracy 77%).
- **Flood frequency:** 4 candidate distributions fit to 30+ years of observed annual peak discharge at each station, cross-checked with 3 goodness-of-fit tests.
- **Uncertainty quantification:** a purpose-built, open-source Python toolkit separates climate-only, land-use-only, and combined effects, and decomposes variance by source (ω^2 -based ANOVA).

2 Model performance

The calibrated hydrological model reproduces observed discharge with satisfactory-to-very-good skill at all three stations: daily Nash–Sutcliffe Efficiency of 0.67–0.77, improving to 0.80–0.93 at the monthly time step. A small, systematic bias emerges and widens moving downstream ($\approx 0\%$ at the upstream station to $\approx -7\%$ at the downstream station) — flagged explicitly for any application sensitive to absolute flow volumes, rather than smoothed over.

3 Key findings

01 Monsoon intensification dominates the climate signal — not directional uncertainty.

Across the 8-member ensemble, temperature increases by +1.6°C to +6.2°C depending on emissions pathway, and precipitation changes range from –2% to +57%. Every "dry" corner still stays close to neutral — the real uncertainty is

how much wetter the monsoon gets, not whether the basin trends wetter or drier.

02 Land-use change has already happened, mostly.

Forest cover rose from 61.0% to 66.8% of the basin area between 2000 and 2020 (cropland fell from 25.0% to 20.9%); the land-use projection framework shows this trend converging toward equilibrium after ~2030, meaning most of the century's land-cover-driven hydrological change is already in the historical record, not still ahead.

03 Which uncertainty source matters depends entirely on the season.

At the representative mid-basin station, climate-model structure alone explains 82–89% of monsoon-season flow-projection variance — but in the dry season and shoulder months, land-use assumptions and climate-model × emissions-pathway interaction effects dominate instead. A single "climate uncertainty" number would misrepresent both halves of the year.

04 All 24 integrated future scenarios agree on the annual direction — not the magnitude.

Every evaluated combination of GCM, emissions pathway, and land-use scenario projects an **increase** in mean annual flow relative to the historical baseline, ranging from **+17%** to **+239%** — a spread of more than 14×. The direction is not in question; the magnitude very much is.

+17% Projected change in mean annual flow, all 24 future scenarios +239%



Season-dependent risk hidden inside a positive annual mean

In the pre-monsoon months, fewer than half of individual ensemble members project flow above the historical baseline — even though the ensemble *mean* for those months stays positive. In the most extreme case (far-future, one month), even the ensemble mean itself turns negative. An annual-scale increase does not guarantee a seasonal one — this is exactly the kind of risk a mean-only summary would hide.

5 Methodological contribution

Beyond the basin-specific results, the project produced a reusable, open-source Python toolkit for uncertainty quantification and source attribution in climate-hydrology impact studies: variance-based decomposition (GCM × emissions pathway × land-use, bias-corrected ω^2), scenario-differencing source attribution (climate-only / land-use-only / combined / interaction effects), probability-of-exceedance framing alongside ensemble means, and flow-duration-curve comparison across scenario types — designed to be channel-agnostic and reusable for other basins and studies. Code, pipeline, and documentation are published on GitHub.

Climate
SWAT+

Hydrology
CMIP6

Uncertainty Quantification

Full methodology and open-source code: github.com/BijayRaj810/streamflow-uncertainty-climate-landuse